

Fuels

A fuel is a substance that burns readily and releases useful energy in the form of heat. Most fuels are carbon compounds. All fuels need to be handled with care so they do not burn too fast. Uncontrollably fast combustions create explosions in which large amounts of energy are released in a very short time.



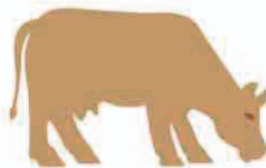
△ Wood

Probably the first fuel used by humans, wood is largely cellulose, made from carbon molecules. Most dried wood burns at about 300°C (572°F), although some types get a lot hotter than this. Other materials are released as smoke when wood burns.



△ Coal

Coal is a flammable rock made from the remains of ancient trees exposed to pressure and heat over time. Its main constituent is pure carbon, although there are many other impurities, including sulfur. Most coals can burn at about 700°C (1,292°F).



△ Methane

Methane, or natural gas, is a simple gas made from hydrocarbons (see page 158). It is found in underground gas fields. It is also produced by natural processes in marshy areas and in the stomachs of herbivores. Its abundance makes it a very popular fuel.



△ Gasoline

Gasoline is a flammable liquid made from hydrocarbons, chiefly octane (C_8H_{18}). It is refined from crude oil (see page 157). The liquid is easy to store in tanks and pump around. It also ignites more easily than other fuels, even from its fumes.



△ Paraffin wax

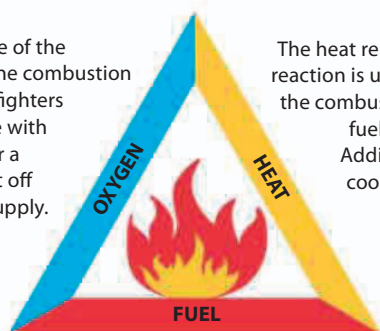
Paraffin wax is a solid and is also made from refined crude oil. The solid does not burn easily, but when melted the liquid wax will ignite. Once lit, the process is self-sustaining—the heat of the combustion melts more of the solid into flammable liquid.

Fire control

Firefighters tackle fires using an understanding of combustion reactions. The fire triangle is a simple way of expressing the three things needed for combustion to continue: oxygen, heat, and fuel. Taking one of these components away will make the reaction end—and the fire go out.

▷ Oxygen

This gas is one of the reactants in the combustion reaction. Firefighters smother a fire with foam, sand, or a blanket to cut off the oxygen supply.



◁ Heat

The heat released by the reaction is used to power the combustion of more fuel and oxygen. Adding water will cool the reaction and reduce its energy.

△ Fuel

Fire needs fuel. Firefighters have to consider what is burning before deciding how best to extinguish the fire safely and effectively.

▽ Fire extinguishers

Different fire extinguishers are designed to tackle fires fueled by different types of substances. For example, water is not used on burning liquid because the hot fuel bubbles up through it, making the water boil and spray the burning fuel into the air.

Type of fire	Fire extinguishers			
	Water	Foam	CO ₂	Powder
 paper, wood, textiles, and plastics	✓	✓		✓
 flammable liquids		✓	✓	✓
 flammable gases				✓
 electrical equipment			✓	✓